Product Matching & Recommendation Ranking Logic

AID-LOG-009 | Revision 1

1. Purpose

This standard defines how AIDOOi determines product eligibility, calculates product-level Match Scores, ranks eligible products, and controls the minimum quality required for recommendation.

2. Core Architecture

Hard requirements determine eligibility. Soft preferences determine ranking. A product must first pass all mandatory eligibility checks. Only eligible products enter Match Scoring.

Stage	System Action
1. Eligibility	Apply Product Type, mandatory practical requirements, fit constraints, availability requirements, and applicable budget eligibility.
2. Match Scoring	Calculate the weighted AIDOOi Match Score for eligible products.
3. Threshold	Exclude products with final Match Score below 60/100.
4. Ranking	Rank remaining candidates from highest to lowest Match Score.
5. Tie-break	If final scores are equal, apply: Fit → Style → Availability → Value.

3. Match Score Formula

Factor	Weight
Fit Quality	30%
Style Match	25%
Colour Match	15%
Budget / Value Match	15%
Practical Match	10%
Availability Convenience	5%
Total	100%

AIDOOi Match Score = Fit 30% + Style 25% + Colour 15% + Budget/Value 15% + Practical 10% + Availability 5%.

4. Fit Quality - 30%

Fit Score	Definition
100	Ideal Fit - highly appropriate scale and placement with desirable circulation.
85	Very Good Fit - strong fit with only minor difference from ideal scale.
70	Acceptable Fit - usable without functional conflict, but scale is not ideal.
50	Weak Fit - technically usable, but space may feel overfilled or under-scaled.
0 / Reject	Fails mandatory spatial, circulation, or functional fit requirements.

Fit uses Room Dimensions, Available Placement Space, Product Dimensions, Required Circulation and Composition Context. Fit is not based on room area alone. For Studio Apartments it is evaluated after the Sleeping Footprint; for Full Apartments and Villas it is evaluated inside the relevant Room Module.

Minimum Fit rule: products below Fit Score 50 are not recommendation candidates.

5. Style Match - 25%

Relationship	Score
Product Primary Style = Customer Style	100
Product Secondary Style = Customer Style	85
Strong / defined compatible style	70
Partially compatible style	50
Clear style conflict	0

Style data follows AID-DATA-009. A controlled Style Compatibility Map must define approved relationships. The system must not rely on uncontrolled AI interpretation for style compatibility. Style mismatch is normally a ranking penalty rather than a hard eligibility failure; clear conflict may be excluded from primary recommendations.

6. Colour Match - 15%

Relationship	Score
Primary Colour directly matches customer direction	100
Secondary Colour directly matches customer direction	85
Compatible Colour Family	70
Neutral / harmonizable with selected palette	50
Clear conflict with Colour Direction	0

Colour data follows AID-DATA-007. Individual Product paths use more direct colour matching. Room and Project paths additionally evaluate Palette Harmony with the evolving composition. A controlled Colour Compatibility Map must govern approved compatible and neutral relationships.

7. Budget / Value Match - 15%

Budget Relationship	Score
Ideal Value Zone	100
Slightly below target	90
Slightly above target but manageable within allocation	75
Significantly below expected level	65
Near maximum tolerable allocation	50
Makes the applicable budget unworkable	Reject

Individual Product pricing uses Current Selling Price and AID-LOG-001. Room and Project paths evaluate the product against its dynamically allocated composition budget, not the total project budget. Lowest price and largest discount are not automatically rewarded; Original Price and discount percentage do not directly increase Match Score.

8. Practical Match - 10%

Practical Relationship	Score
All relevant practical preferences satisfied	100
Almost all satisfied; minor difference	85
Core needs satisfied; some preferred attributes absent	70
Minimum required function satisfied	50
Mandatory practical requirement not satisfied	Reject

Practical requirements are category-specific and reference AID-CAT-001 through AID-CAT-008. Mandatory requirements determine eligibility; optional attributes improve ranking only when relevant to known customer needs or composition context. More features do not automatically mean a better match. This logic must not create unnecessary customer questions.

9. Availability Convenience - 5%

Availability Situation	Score
In Stock + On Display + Ready/Fast Delivery	100
In Stock + Viewable on Request + Ready/Fast Delivery	85

Availability Situation	Score
In Stock + No Immediate Viewing + Available for Delivery	70
Available with longer delivery or operational limitation	50
Unavailable / Inactive	Reject

Availability references AID-DATA-011. Availability determines whether a product can realistically be recommended, while convenience improves ranking without overriding product suitability.

10. Recommendation Thresholds

Final Match Score	Classification
90-100	Excellent Match
80-89	Strong Match
70-79	Good Match
60-69	Acceptable Match
Below 60	Do Not Recommend

Minimum Recommendation Score = 60/100. Passing Hard Filters does not guarantee recommendation. Products below the threshold must not be used merely to fill result slots. If suitable inventory is insufficient, AIDOOi should surface limited matching inventory rather than lower recommendation quality.

11. Tie-breaker Logic

When eligible products have the same final Match Score, rank them by: 1. Higher Fit Score → 2. Higher Style Score → 3. Higher Availability Score → 4. Higher Budget/Value Score. If still equal, candidates may remain equivalent until a later composition-level rule resolves the choice.

12. Controlled Compatibility Maps

Style and Colour compatibility must be governed by controlled maps maintained by AIDOOi. The maps may evolve over time, but production ranking must use approved relationships rather than unconstrained model interpretation. This preserves consistency, explainability and testability.

13. Scope Boundary

AID-LOG-009 ranks individual eligible products. It does not define the four final customer-facing recommendation compositions. That responsibility belongs to AID-LOG-010 - Final Recommendation Options Logic. Room/project visualization is governed separately by the future AID-VIS-001 standard.

14. Final Principle

AIDOOi must never allow a strong soft-preference score to compensate for a failed mandatory requirement. Eligibility protects functional correctness; Match Scoring determines preference quality; the recommendation threshold protects overall output quality.